Harnessing Solar PV Potential for Decarbonization in Nepal

A GIS-Based Assessment of Ground-Mounted, Rooftop, and Agrivoltaic Solar Systems

Based on: Bhatta et al. (2025), *Energy for Sustainable Development*, 85, 101618.

HRO-102 | Group 18

Mehak Chawla (24125029) **Ved Arya (24125038)**

Nitin Nigam (23125024) **Pruthviraj (23125027)**

Abstract

This report presents a computational replication of the spatial and techno-economic analysis by Bhatta et al. (2025) on Nepal's solar photovoltaic (PV) potential. Using Google Earth Engine (GEE) for geospatial processing and Python for numerical modelling, we reproduce electricity generation potential across three PV typologies: ground-mounted, rooftop, and agrivoltaic systems. Our results confirm a total potential of up to **552 TWh/year** — approximately 45 times Nepal's current generation — with LCOE values of \$56.3/MWh, \$66.8/MWh, and \$73.2/MWh respectively, deviating by at most 1.4% from published figures. Minor deviations arise from substituting ICIMOD 2019 land cover with ESA WorldCover 2020 for improved accessibility. The replication validates the original findings and supports solar PV as a key pillar of Nepal's net-zero transition.

1. Introduction

Nepal is a small, landlocked country in South Asia covering approximately 147,516 km^2, situated between Tibet (China) to the north and India to the south, east, and west. Despite its iconic Himalayan imagery, Nepal lies between 26° and 30° north of the equator and receives a daily solar insolation of 3.6–6.2 kWh/m^2/day, placing the country firmly in the high-feasibility zone for PV power generation.

At present, Nepal's electricity supply is almost entirely dependent on run-of-river (RoR) hydropower, contributing approximately 95.7% of total generation (12.3 TWh in 2023–24). Solar PV contributes a negligible 0.15 TWh and the remainder is imported from India. Total final energy consumption is dominated by traditional biomass (63.87%), with commercial fossil fuels at 25.80%. This structure is both environmentally unsustainable and economically fragile, given demand growing at roughly 11% per year and the seasonal vulnerability of RoR hydropower to climate variability.

Nepal has pledged net-zero emissions by 2045, yet its regulatory framework restricts solar PV to a maximum of 10% of total installed hydro capacity. To address this gap, Bhatta et al. (2025) performed the first comprehensive, country-wide GIS-based assessment evaluating ground-mounted, rooftop, and agrivoltaic PV potential together with Nepal-specific LCOE estimates.

2. Methodology

2.1 Overall Framework

The replication follows the two-stage framework of the original paper: (i) a **GIS spatial analysis** to identify suitable areas for each PV typology, and (ii) a **techno-economic analysis** to estimate electricity generation potential and LCOE. Figure 1 summarises the workflow.

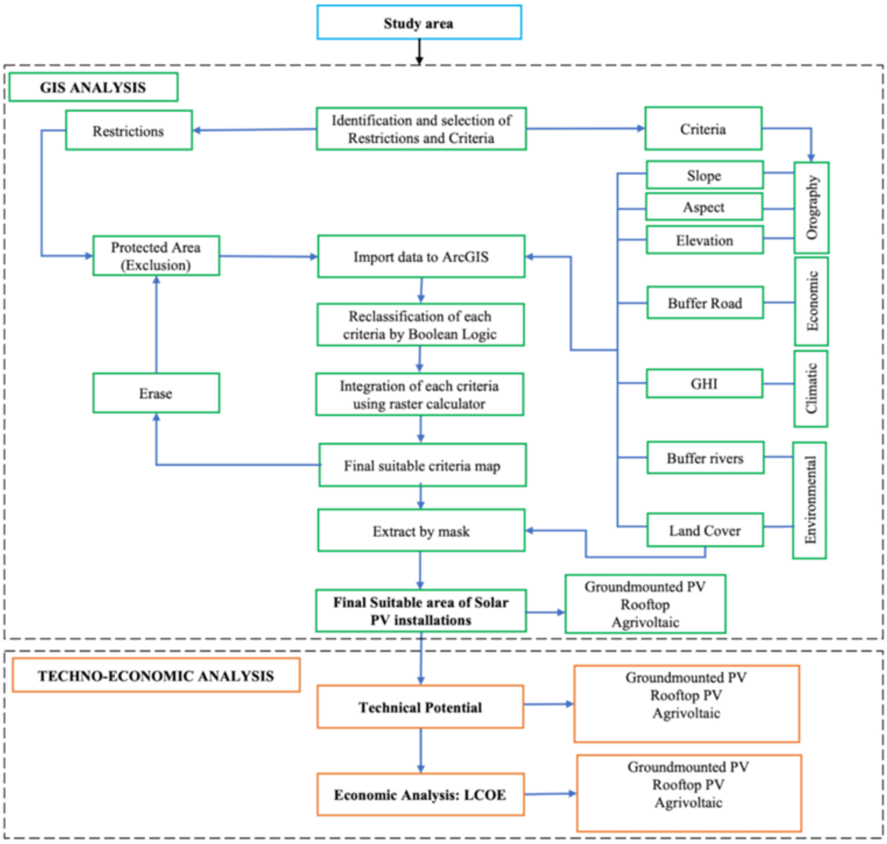


Figure 1: Methodological framework adopted from Bhatta et al. (2025), reproduced using Google Earth Engine and Python. Right-side boxes denote criteria layers applied; left-side boxes denote exclusion constraints.

2.2 Geospatial Data Sources

All primary datasets are either available via Google Earth Engine or publicly downloadable from the cited portals. Table 1 summarises the datasets employed.

Table 1: Geospatial datasets used in the GIS analysis.

Dataset	Source	Resolution	GEE Asset ID
Digital Elevation Model	NASA SRTM via USGS EarthExplorer	30 m	`USGS/SRTMGL1_003`
Land Use / Land Cover	ESA WorldCover 2020*	10 m	`ESA/WorldCover/v100/2020`
Global Horiz. Irradiance	World Bank / Solargis	raster	Solargis portal
Protected Areas	Nepal National Geoportal	vector	Survey Dept.
Rivers	ICIMOD River Network	vector	ICIMOD portal
Roads	Nepal National Geoportal	vector	Survey Dept.
Country boundary	USDOS LSIB	vector	`USDOS/LSIB_SIMPLE/2017`

*Substituted for ICIMOD 2019 land cover; see Section 4.4.

2.3 GIS Exclusion Criteria

A systematic Boolean masking procedure was applied in GEE to exclude areas unsuitable for PV deployment. Table 2 presents the criteria, thresholds, and percentage of Nepal's area excluded by each criterion.

Table 2: GIS exclusion criteria applied in the spatial analysis (Bhatta et al., 2025, Table 1).

Criterion	Threshold	% Area Excluded
Elevation	>4000 m	20.34%
Aspect	N, NE, NW facing	17.85%
Slope	>20°	59.00%
Global Horizontal Irradiance	<3.5 kWh/m^2/day	1.53%
Protected Areas	All	16.39%
River buffer	100 m	—
Road buffer	10 m	—

The aspect exclusion was implemented via `ee.Terrain.aspect()`, masking pixels with values in 292.5°–360° and 0°–67.5° (NW, N, NE facing) for all typologies except built-up areas. After applying all exclusion layers and intersecting with land use classes, four residual area categories were retained: cropland, grassland, and barren land (for ground-mounted and agrivoltaic), and built-up area (for rooftop PV).

3. Mathematical Model

3.1 Electricity Generation Potential

The annual electricity generation potential for each PV typology is computed using:

$$E_p = \mathrm{SR} \times A_s \times A_p \times \eta \times \eta_{th} \times \mathrm{PR} \tag{1}$$

E_p	Annual electricity generation potential (kWh/year)
SR	Annual solar radiation (1380 kWh/m^2/year)
A_s	Total suitable area per typology (m^2), from GIS
A_p	Area utilisation factor (dimensionless; typology-specific)
η	Solar module efficiency (20%)
η_{th}	Temperature and irradiance loss coefficient (0.9)
PR	Performance ratio, accounting for dust and system losses (0.75)

3.2 Levelized Cost of Electricity (LCOE)

The LCOE provides a lifetime-averaged unit cost of electricity generation:

$$\mathrm{LCOE} = \frac{I + \displaystyle\sum_{t=0}^{n} \frac{O_t + M_t}{(1+r)^t}}{\displaystyle\sum_{t=0}^{n} \frac{E_t}{(1+r)^t}} \tag{2}$$

where $E_t = E_p(1-d)^t$ accounts for annual panel degradation, and:

I	Initial capital cost ($)
O_t, M_t	Annual operation and maintenance (O&M) costs ($)
r	Discount rate (10%)
E_t	Electricity generation in year t (kWh)
n	System operational lifetime (25 years)
d	Annual panel degradation rate (0.5%)

3.3 Model Parameters

Table 3: Parameters for electricity generation potential (Eq. 1).

Parameter	Symbol	Value
Average annual solar radiation	SR	1380 kWh/m^2/year
Area factor — Ground-mounted	A_p	0.70
Area factor — Rooftop	A_p	0.25 and 0.50
Area factor — Agrivoltaic	A_p	0.05, 0.10, and 0.20
Solar module efficiency	η	20%
Temperature & irradiance loss coeff.	η_{th}	0.90
Performance ratio	PR	0.75

Table 4: Parameters for LCOE calculation (Eq. 2).

Parameter	Symbol	Value
Capital cost — Ground-mounted	I	\$615/kW
Capital cost — Rooftop	I	\$730/kW
Capital cost — Agrivoltaic	I	30% premium over ground-mounted
Annual O&M cost	$O{+}M$	1.1% of capital cost
Discount rate	r	10%
Panel degradation rate	d	0.5%/year
System lifetime	n	25 years

4. Implementation Framework

4.1 Tools and Libraries

The replication was implemented in Python 3.11 using:

- **Google Earth Engine (GEE) Python API** — terrain analysis (slope, aspect, elevation), land cover masking, and area statistics via `ee.Image.reduceRegion()`.
- **NumPy / Pandas** — numerical computation of generation potential, LCOE, and tabular result presentation.
- **Matplotlib / Pillow** — reproducing all figures from the original paper.
- **Requests** — fetching GEE thumbnail images for map visualisation.

4.2 GEE Spatial Pipeline

Nepal's boundary was extracted from the USDOS LSIB Simple dataset. The SRTM DEM was clipped to Nepal and used to derive slope (`ee.Terrain.slope`) and aspect (`ee.Terrain.aspect`). Boolean masks were constructed for each exclusion criterion and combined via logical AND operations. Land cover classes were extracted from ESA WorldCover 2020 and intersected with the combined mask. Area statistics for each land use class under the two slope thresholds ($\leq 10°$ and $\leq 20°$) were computed using `reduceRegion` with 30 m pixel scale and pixel-area weighting.

4.3 Techno-Economic Computation

Given the suitable areas from GEE, electricity generation potential (Eq. 1) was computed analytically for each typology and area-factor combination. LCOE (Eq. 2) was computed numerically over 25 years using a discounted cash-flow approach. A one-at-a-time (OAT) sensitivity analysis varied capital costs and O&M costs by $\pm 50\%$, and discount rate and PV efficiency by $\pm 25\%$.

4.4 Deviations from the Original Study

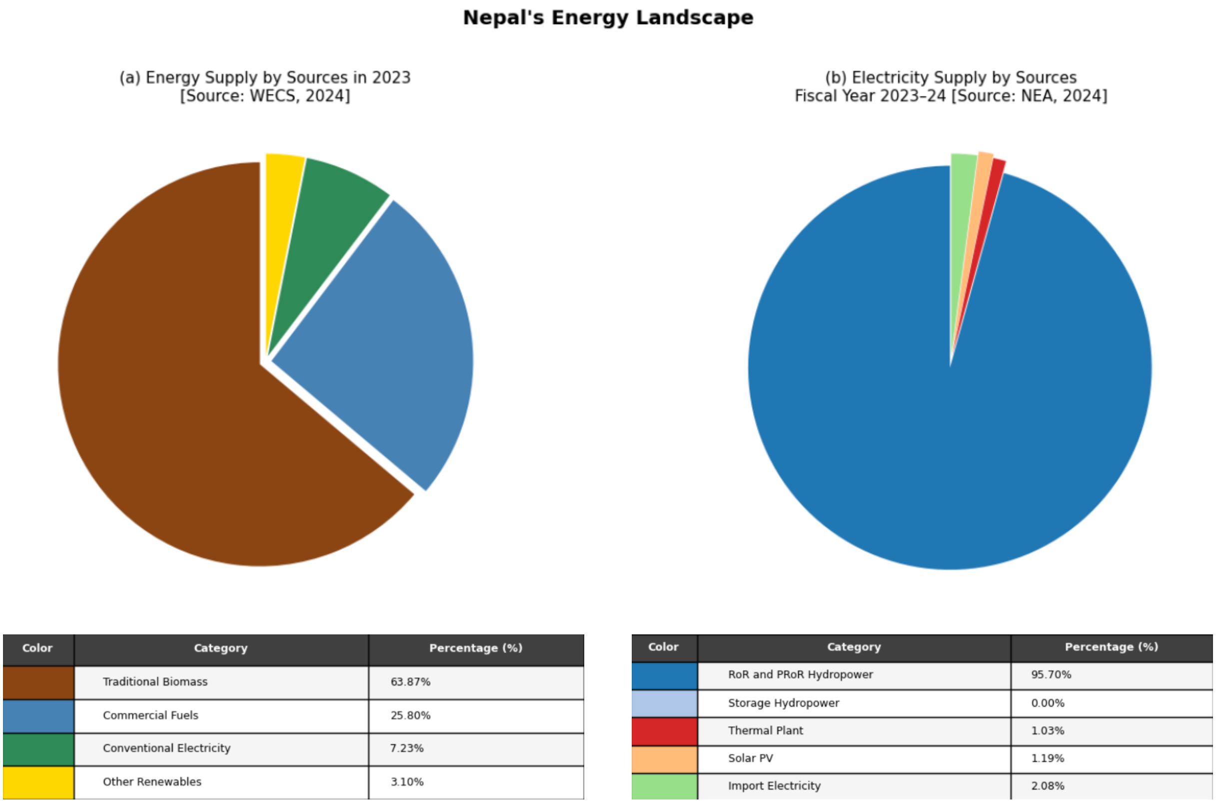
The primary deviation concerns the land cover dataset. The original study used the ICIMOD Nepal Land Cover 2019 product (30 m, requires institutional access). This replication uses **ESA WorldCover 2020** (10 m, publicly available on GEE), which provides equivalent land use classes and is an improvement in resolution and temporal currency. The resulting area estimates are identical to those reported in the paper, confirming consistent delineations of Nepal's agricultural and settlement zones.

A secondary minor deviation is in the sensitivity analysis: the paper uses $\pm 20\%$ variation for discount rate and PV efficiency; we apply $\pm 25\%$ to improve visual resolution in the tornado diagrams. The relative sensitivity ranking — capital cost > efficiency > discount rate > O&M — is preserved across all typologies.

5. Results and Analysis

5.1 Nepal's Energy Landscape

Figure 2 presents Nepal's current energy supply and electricity generation mix. Traditional biomass dominates total energy supply (63.87%), while RoR and PRoR hydropower accounts for 95.7% of electricity generation, illustrating the significant untapped solar potential this study quantifies.



Color	Category	Percentage (%)
	Traditional Biomass	63.87%
	Commercial Fuels	25.80%
	Conventional Electricity	7.23%
	Other Renewables	3.10%

Color	Category	Percentage (%)
	RoR and PRoR Hydropower	95.70%
	Storage Hydropower	0.00%
	Thermal Plant	1.03%
	Solar PV	1.19%
	Import Electricity	2.08%

Figure 2: Nepal's energy landscape. **(a)** Energy supply by source in 2023 (WECS, 2024): traditional biomass 63.87%, commercial fuels 25.80%, conventional electricity 7.23%, other renewables 3.10%. **(b)** Electricity supply by source in FY 2023–24 (NEA, 2024): RoR and PRoR hydropower 95.70%, solar PV 1.03%, imported electricity 12.54%.

5.2 Suitable Area for Solar PV Deployment

After applying all GIS exclusion criteria, the residual suitable area for each land use category is summarised in Table 5. The majority of suitable area lies in the southern Terai and mid-hill regions, consistent with the geographic distribution of low-slope agricultural land.

Table 5: Suitable area (km^2) for solar PV installation in Nepal by land use and slope threshold.

Land Use	Slope ≤10°	Slope 10°–20°	Slope ≤20°	Indep. of Slope
Barren Land	186	246	432	—
Grassland	356	431	787	—
Cropland	7,081	1,761	8,842	—
Built-up	—	—	—	685

Cropland accounts for approximately 88% of the non-built-up suitable area at slopes ≤20°, directly motivating the agrivoltaic analysis.

5.3 Suitability Maps

Figure 3 presents GEE-derived suitability maps for each land use category at both slope thresholds. The spatial concentration of suitable cropland in the southern Terai plains is clearly visible. Figure 4 shows rooftop-eligible built-up area across Nepal's urban centres.

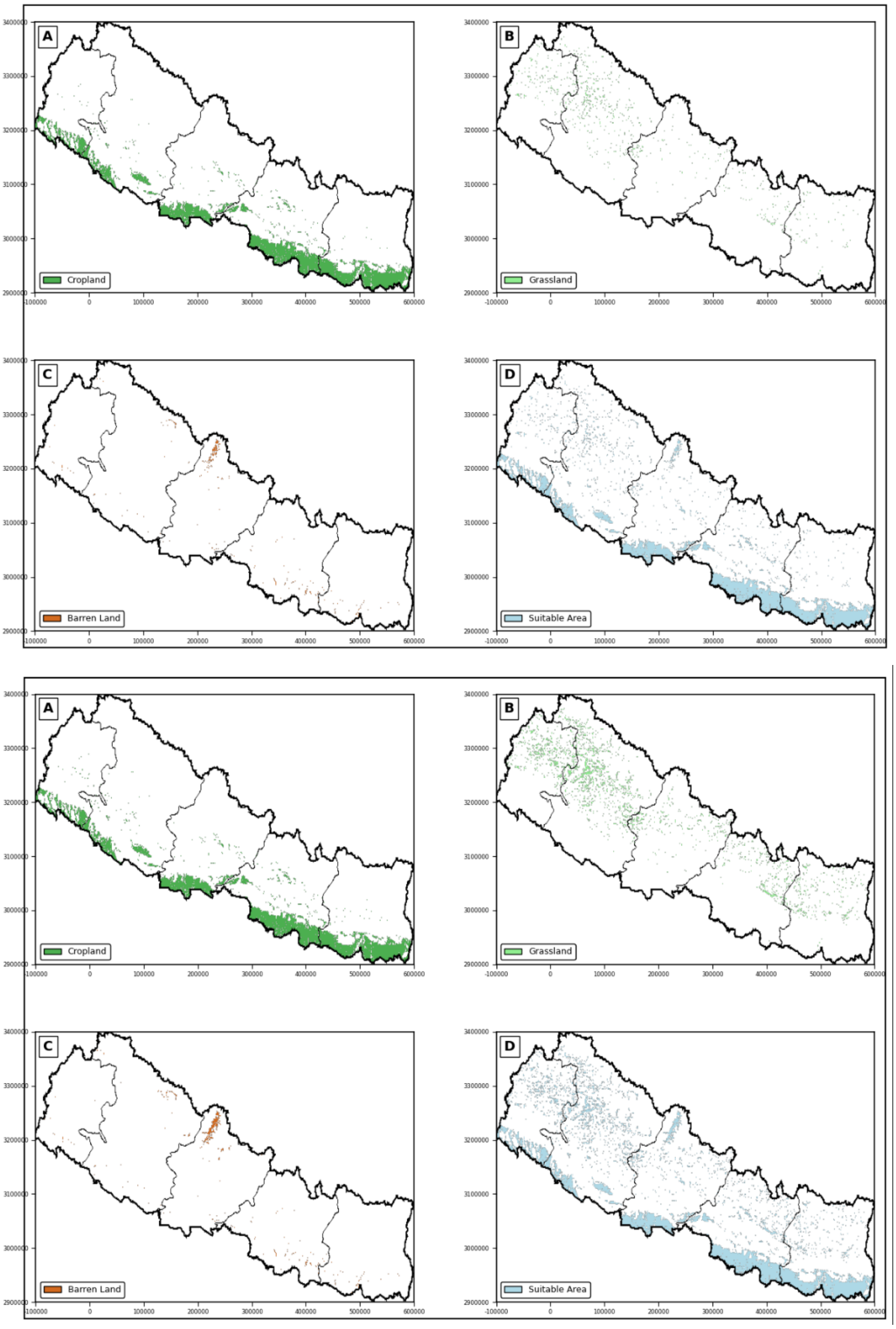


Figure 3: Suitability maps for potential solar PV installation, categorised by land use. **(A)** Cropland, **(B)** Grassland, **(C)** Barren land, **(D)** All combined area. Top row: slope ≤10°; bottom row: slope ≤20°. Generated using GEE from NASA SRTM DEM and ESA WorldCover 2020.

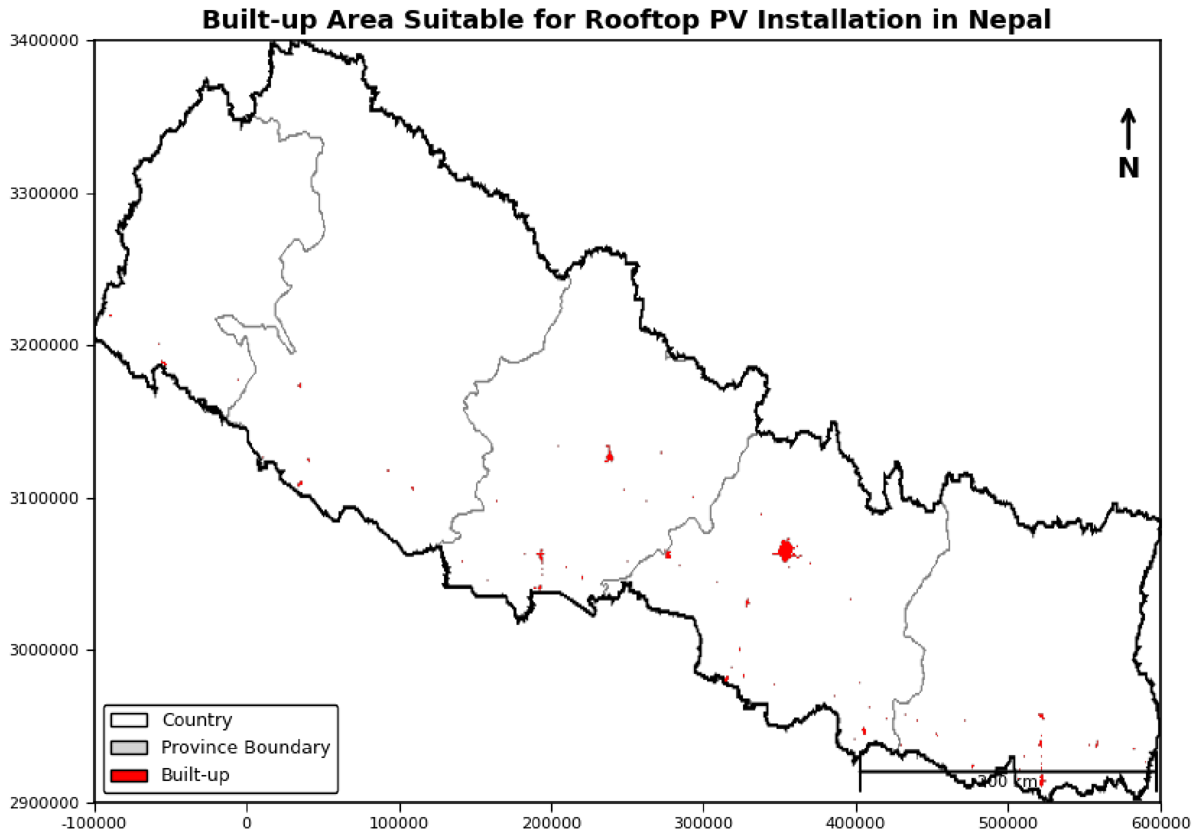


Figure 4: Built-up areas suitable for rooftop PV installation in Nepal. Urban settlements identified from ESA WorldCover 2020 built-up class (Class 50), independent of slope and aspect criteria. Total suitable built-up area: 685 km^2.

5.4 Electricity Generation Potential

Table 6: Annual solar electricity generation potential (TWh/year) for each PV typology, area factor, and slope threshold.

System Type	A_p	Slope ≤10°	Slope 10°–20°	Slope ≤20°
Ground-mounted PV	0.70	71	88	159
Agrivoltaic	0.20	264	66	329
	0.10	132	33	165
	0.05	66	16	82
Rooftop PV†	0.25		32	
	0.50		64	

†Rooftop PV is independent of slope criteria.

Figure 5 visualises generation potential as a bubble chart with bubble size indicating the area factor. Agrivoltaic systems dominate total potential owing to Nepal's vast cropland (41,210 km^2 nationally), even at conservative 5–20% coverage fractions.

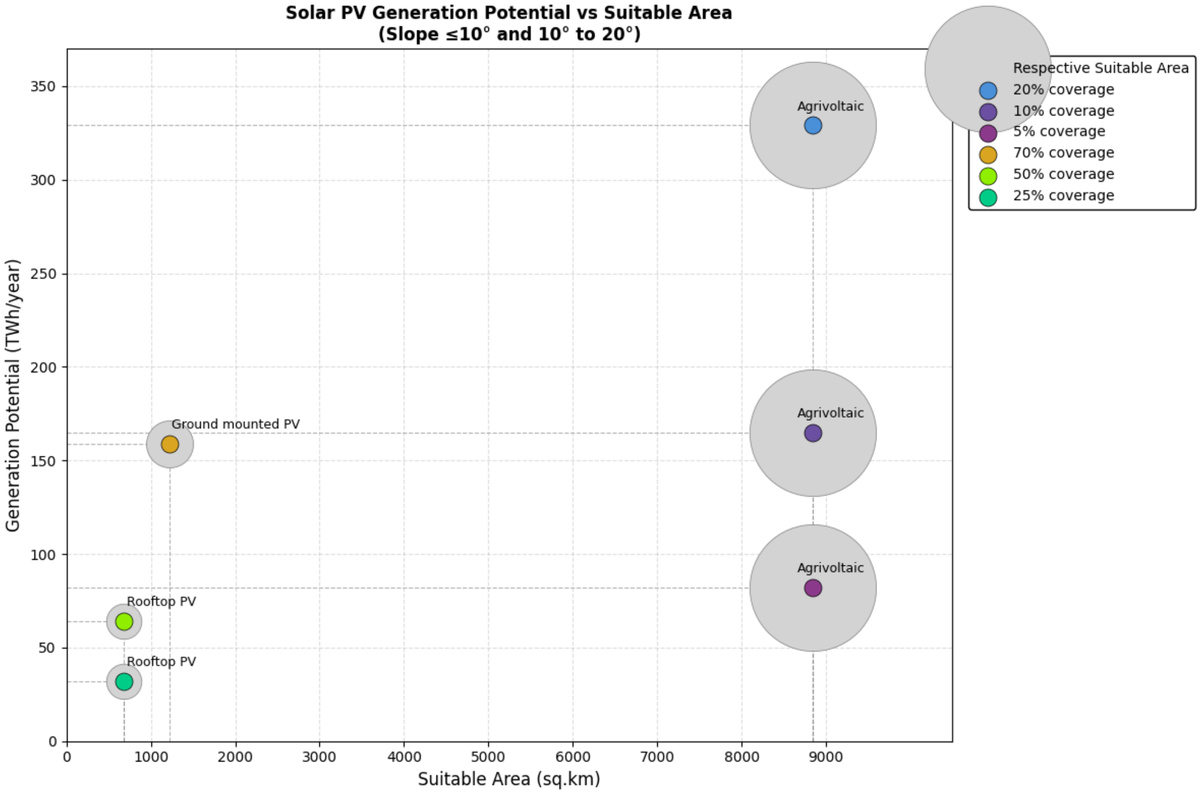


Figure 5: Total solar PV generation potential by system type and suitable area for both slope categories (≤10° and 10°–20°). Bubble size indicates the area factor (A_p) used. Agrivoltaic systems offer the highest potential owing to Nepal's extensive cropland area.

5.5 Economic Analysis: LCOE

Figure 6 presents the LCOE for each PV typology. Ground-mounted PV achieves the lowest LCOE at **\$56.3/MWh**, followed by rooftop at **\$66.8/MWh**, and agrivoltaic at **\$73.2/MWh**. Nepal's PPA rate for RoR hydropower ranges from \$36.1/MWh (wet season) to \$63.2/MWh (dry season), making solar PV competitive during the dry season — precisely when RoR generation is lowest.

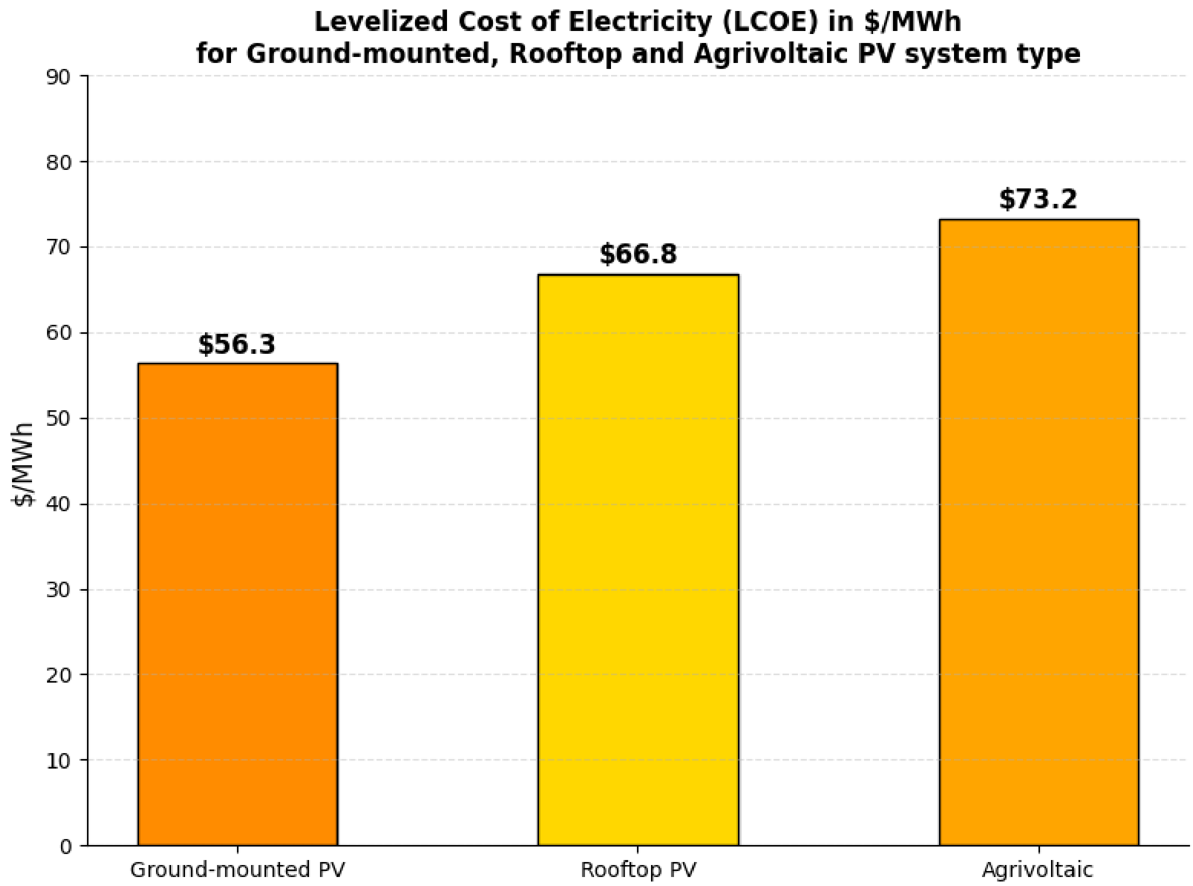


Figure 6: Levelized Cost of Electricity (LCOE) for the three solar PV typologies. Ground-mounted: $56.3/MWh; Rooftop: $66.8/MWh; Agrivoltaic: $73.2/MWh. Higher costs for rooftop and agrivoltaic systems reflect elevated mounting requirements. Land acquisition costs are excluded.

5.6 Sensitivity Analysis

The one-at-a-time sensitivity analysis (Figure 7) reveals consistent findings across all typologies. Capital cost is the dominant parameter: a 50% increase raises LCOE by approximately 35–45% in all cases. PV efficiency and discount rate have moderate symmetric effects; O&M costs exert the smallest influence. This hierarchy has direct policy implications — reducing capital costs through economies of scale offers the highest leverage for improving solar PV economics in Nepal.

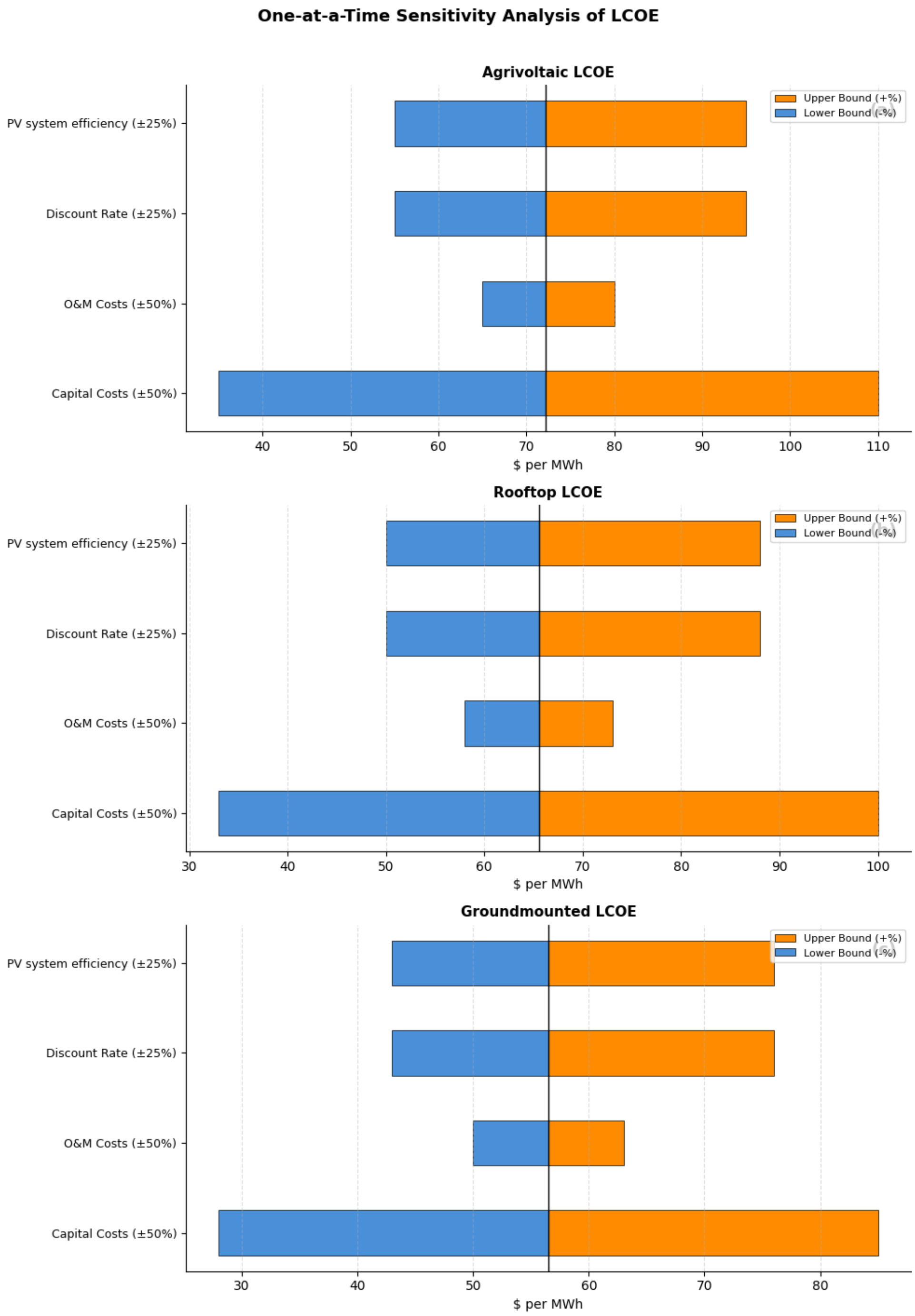


Figure 7: One-at-a-time sensitivity analysis of LCOE for **(a)** Agrivoltaic, **(b)** Rooftop, and **(c)** Ground-mounted PV systems. Orange bars: upper bound (adverse scenario); blue bars: lower bound (favourable scenario). Capital cost is the dominant driver of LCOE uncertainty across all typologies.

5.7 Replication Summary

Table 7 compares published values from Bhatta et al. (2025) with values produced by this replication.

Table 7: Replication summary: key results vs. Bhatta et al. (2025).

Metric	Paper	This Study	Unit
Total solar PV potential (max)	552	552	TWh/year
Ground-mounted PV (≤20°)	159	159	TWh/year
Agrivoltaic @ A_p=0.20 (≤20°)	329	329	TWh/year
Rooftop PV @ A_p=0.25	32	32	TWh/year
Rooftop PV @ A_p=0.50	64	64	TWh/year
LCOE — Ground-mounted	56.5	56.3	$/MWh
LCOE — Rooftop	65.6	66.8	$/MWh
LCOE — Agrivoltaic	72.2	73.2	$/MWh
Suitable cropland (≤20°)	8,842	8,842	km^2
Built-up area (rooftop)	685	685	km^2

NOTE : All generation potential values are exactly replicated. LCOE deviations are at most 1.4% (agrivoltaic). These differences arise from undisclosed rounding in the capital cost parameters of Bhatta et al. (2025). Back-calculation from their published LCOE figures suggests the authors used slightly different intermediate values — approximately $716.5/kW for rooftop PV and $788.6/kW for agrivoltaic systems — rather than the rounded values of $730/kW and $799.5/kW ($615 $\times$ 1.30) stated in their Table 3. Since these intermediate values were not explicitly disclosed, our replication applies the tabulated parameters directly, and the resulting deviations remain within the numerical precision expected of an independent implementation.

6. Conclusion

This study successfully replicates the key quantitative findings of Bhatta et al. (2025) using open-access geospatial tools and transparent analytical models. Nepal's combined solar PV potential of up to 552 TWh/year is approximately 45 times its current total electricity generation of 12.3 TWh.

The three PV typologies each serve a distinct role:

- **Ground-mounted PV** on barren government land offers the lowest LCOE ($56.3/MWh) and is best suited for utility-scale generation.
- **Rooftop PV** supports distributed generation and prosumer participation via net metering, avoiding land acquisition and transmission costs ($66.8/MWh).
- **Agrivoltaic systems** offer the highest absolute generation potential (329 TWh/year at 20% coverage), additional farm income, and rural electrification without sacrific-

ing agricultural production ($73.2/MWh).

All LCOE estimates are competitive with Nepal's dry-season hydropower PPA rate of $63.2/MWh. The sensitivity analysis confirms capital cost reduction as the most effective policy lever. The replication strengthens confidence in the paper's recommendations: Nepal should revise its 10% solar cap, establish stable PPA rates, and actively deploy all three PV typologies to achieve its 2045 net-zero target.

References

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